

Project Update

Numerical Simulation of Manufacturing Process

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- ① Progress
- ② Currently working on
- ③ Future work

① Progress

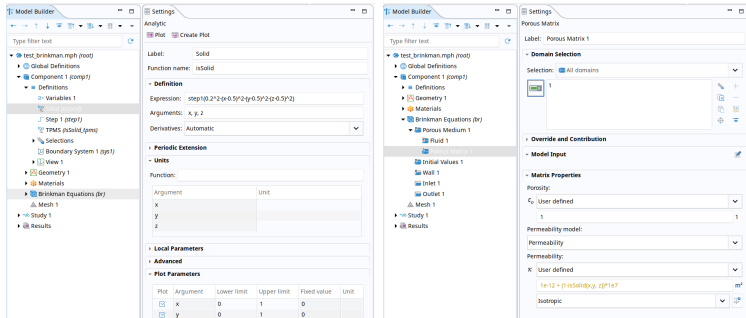
② Currently working on

③ Future work

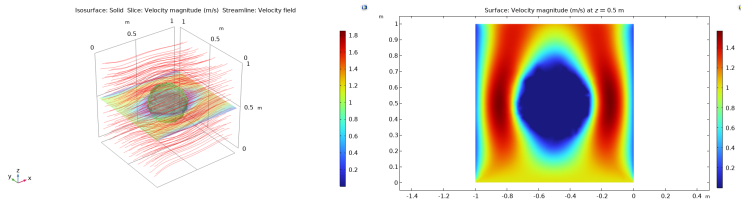
Sphere example

Before considering a TPMS structure, we have to be familiar with Comsol and Brinkman penalization. For Axel, this means starting with a more simple example.

- Define an analytic function for the obstacle
- Define the porous medium properties for the region



Sphere example results



Conclusion: Brinkman frameworks works for very simple geometries

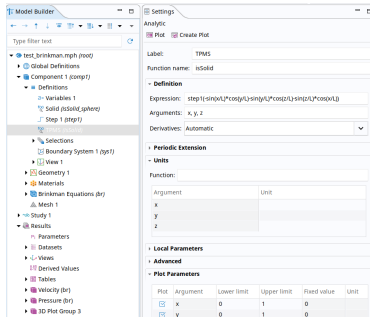
TPMS analysis

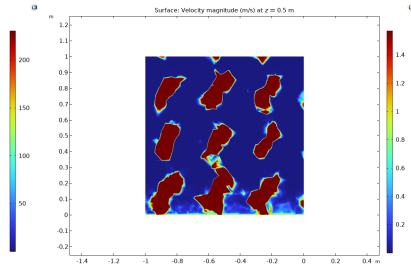
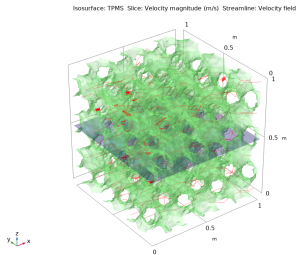
The gyroid TPMS structure can be implemented in the same way as the sphere, i.e. by an analytic expression.

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- Incorporate heat transfer into the model
- Successfully implement a two-channel model